

SKETCHING ALGORITHMS

- ▶ "limited" space to answer queries
- ▶ max = easy to compute on a stream of sets using $O(1)$ words of memory

if $\text{current max} < \text{current item}$
then $\text{current max} \leftarrow \text{current item}$

TAKE NEXT ITEM

- ▶ Toy problem: having knowledge of an integer n
find the missing item in a stream of $n-1$ different integers from $[1..n]$

example: $n = 5$ stream: $\boxed{5} \boxed{3} \boxed{1} \boxed{2} \rightarrow$ missing item is $\boxed{4}$

a) Inefficient way to solve the problem:

keep a binary vector B of n bits initially set to $\emptyset$

 $B[\text{current item}] \leftarrow 1$

TAKE NEXT ITEM

Scan B and find the position i s.t. $B[i] = 0$

Output i as missing item

b) Better way: use Gauss' sum

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} = S$$

 $S \leftarrow S - \text{current item}$

TAKE NEXT ITEM

output S

example: $n = 5$ stream: $\boxed{5} \boxed{3} \boxed{1} \boxed{2} \rightarrow$ missing item is $\boxed{4}$

$$S = \frac{5 \cdot 6}{2} = 15 \rightarrow 10 \vee 7 \vee 6 \vee 2 \rightarrow 4$$

How many bits? We need to store S and it requires $\log_2\left(\frac{n(n+1)}{2}\right)$
bits $\sim 2 \log n$ bits

c) Can we use $\log n$ bits?

XOR: associative, commutative, $\text{XOR}^{-1} = \text{XOR}$

$$S \leftarrow \text{XOR}_{i=1}^n i \quad (\text{in binary})$$

$S \leftarrow S \text{ XOR current item}$

TAKE NEXT ITEM

OUTPUT S

now S takes $\log n$ bits

► A stream $a_1, a_2, \dots, a_m$ is given

We say that item a has frequency $f_a = |\{a_j \mid 1 \leq j \leq m, a_j = a\}|$

example: 1 3 1 5 2 1 3 3 2 2 2 5 1 4 3 2

$$f_3 = 4$$

Item a is the most frequent if $f_a \geq f_b$ for any other item $b \neq a$

Exercise: Design a streaming algorithm that uses $O(1)$ words and find the most frequent item a when $f_a > \frac{m}{2}$

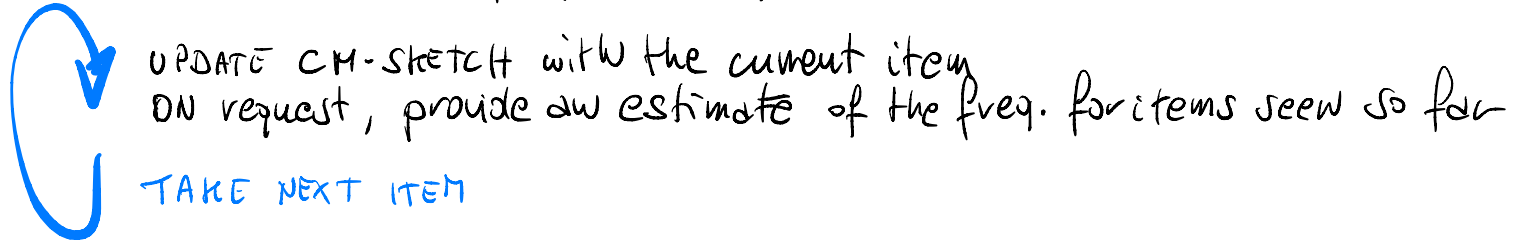
Real world application (even when f_a is not so large) :

Denial of service attack

(we don't see: it cannot be solved deterministically in the general case)

► TODAY'S TOPIC: COUNT-MIN SKETCH (CM-SKETCH)

We want to estimate the frequencies of the items in an unbounded stream



We would like to get something like a FPTAS (ϵ, δ)

approximation ϵ error probability δ

SCENARIO :

- ▶ n distinct items, numbered from 1 to n , in an unbounded stream (n is huge)
- ▶ frequency array F s.t. $F[i] = \#$ times that item i has been seen so far
- ▶ update operation : $F[i]++$ for the current item i
- ▶ query operation for any given item i : return $F[i]$

CHALLENGE : USE AS LITTLE SPACE AS POSSIBLE (polylogarithmic)

Cannot be done deterministically (we do not see this)

$\Rightarrow$ use FPTAS approach (with ϵ, δ)

GOAL : provide an approximation $\tilde{F}$ for F with bounded error probability

For this, let $\|F\| = \sum_{i=1}^n F[i]$ = sum of all counters (recall that we cannot keep F as n is very large)

↑
norm of F

Given item i , we estimate $F[i]$ as $\tilde{F}[i]$ s.t.

$$F[i] \leq \tilde{F}[i] \leq F[i] + \epsilon \|F\|$$

always true
it holds with prob. $\geq 1 - \delta$

Note that ϵ gives an absolute "approximation" rather than a relative one:

ideally we would like to have $\tilde{F}[i] \leq F[i] + \epsilon F[i]$ but this is not possible.

$\hookrightarrow \|F\|$

For example if $F[i] = 0$, we could certainly

establish that i did not appear in the stream

which is a rather strong (and impossible) claim $(\tilde{F}[i] \leq (1+\epsilon)F[i]) \Rightarrow \tilde{F}[i] = 0$

The reason is that we are using an incredible small amount of space

$O(\varepsilon^{-1} \log \delta^{-1})$ words of memory (each of $O(\log \|F\|)$ bits)

(note that CB usually give a factor of ε^{-2} , instead here it is ε^{-1} ... think e.g. when $\varepsilon = 10^{-6}$)

Let's describe how it works

We need universal hash family $\mathcal{H}$ that is 2-wise independent

$$h(x) = (ax + b) \% p \% C$$

Algorithm CM-SKETCH (ε, δ):

1. # rows $r = \lceil \ln S^{-1} \rceil$, # columns $c = \frac{e}{\varepsilon}$ where r, c are the closest integers and $e = 2.71828...$ is the Euler's constant
2. Two-dimensional array T (the table) of size $r \times c$, where each entry is a counter initialized to ϕ .
3. Randomly and uniformly chosen $h_0, h_1, \dots, h_{r-1} \in \{ \}$ (the domain is $[c]$)

OBS

		0	1	2	...						$c-1$
h_0	0										
h_1	1										
h_2	2										
$\vdots$	$\vdots$										
h_{r-1}	$r-1$										

T

SPACE
 $O(rc) = O(\varepsilon^{-1} \ln S^{-1})$
 words

item $i \in [n] \rightsquigarrow$ we associate r entries in T , one per row
 $(0, h_0(i)), (1, h_1(i)), (2, h_2(i)), \dots, (r-1, h_{r-1}(i))$

UPDATE $F[i]++$

$\hookrightarrow T[0][h_0(i)]++ , \dots$

$T[r-1][h_{r-1}(i)]++$

$O(r) = O(\log^{-1})$ time

	0	1	2	...	C-1
$h_0, 0$	++				
$h_1, 1$				++	
$h_2, 2$			++		
$\vdots$					
$\vdots$				++	
$h_{r-1}, r-1$					++

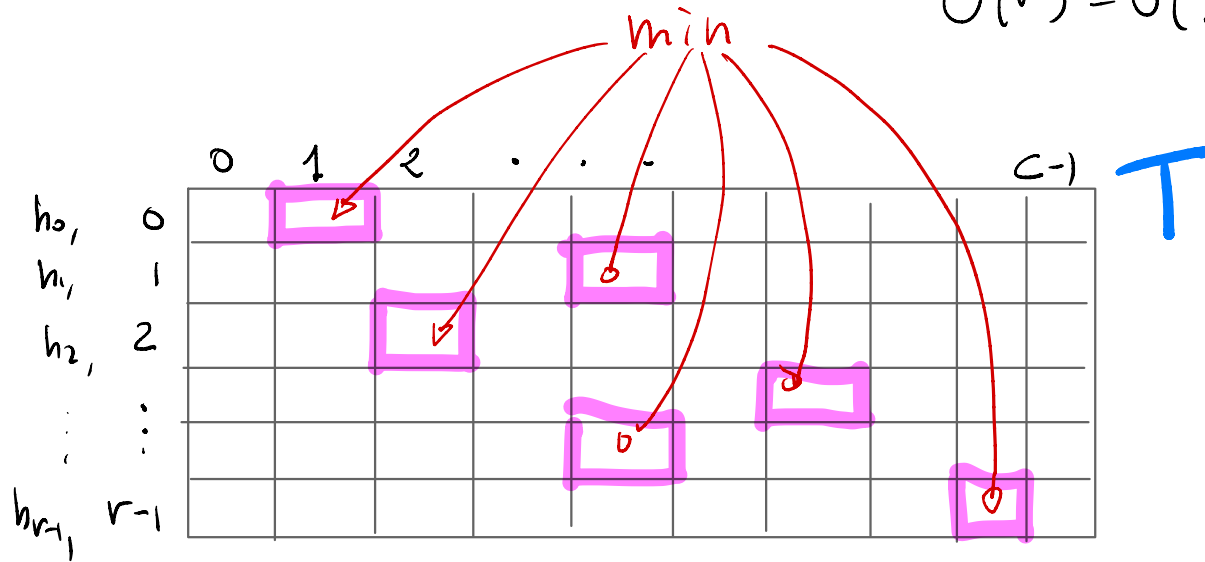
T

item $i \in [n] \rightsquigarrow$ we associate r entries in T , one per row
 $(0, h_0(i)), (1, h_1(i)), (2, h_2(i)), \dots, (r-1, h_{r-1}(i))$

Query for i : $F[i]$

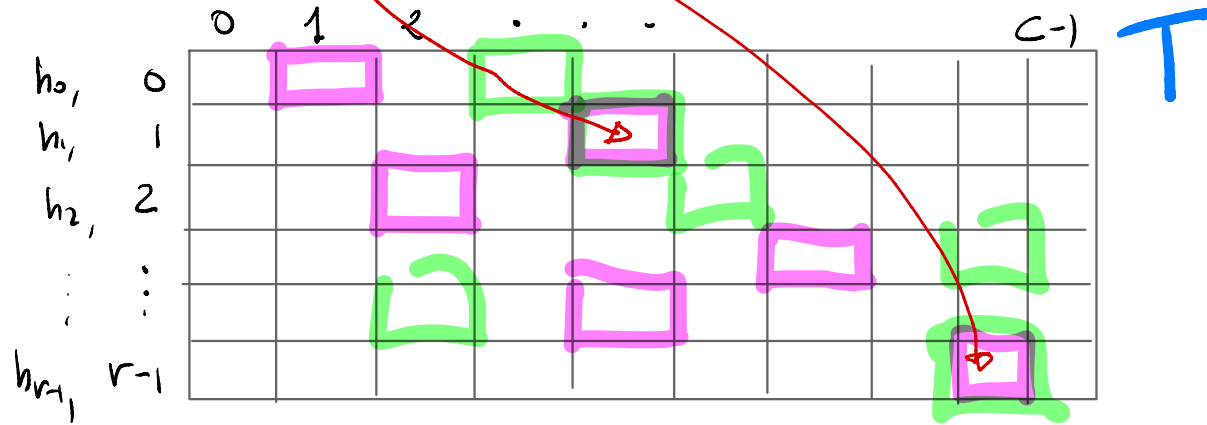
$\hookrightarrow$ return $\tilde{F}[i] = \min \{ T[0][h_0(i)], \dots, T[r-1][h_{r-1}(i)] \}$

$O(r) = O(\lg d^{-1})$ time



item $i \in [n] \leadsto$ we associate r entries in T , one per row
 $(0, h_0(i)), (1, h_1(i)), (2, h_2(i)), \dots, (r-1, h_{r-1}(i))$

Issue: collisions in some entries of T by two distinct items i and i'



item $i \in [n] \rightsquigarrow$ we associate r entries in T , one per row
 $(0, h_0(i)), (1, h_1(i)), (2, h_2(i)), \dots, (r-1, h_{r-1}(i))$

item $i' \in [n] \rightsquigarrow$ we associate r entries in T , one per row
 $(0, h_0(i')), (1, h_1(i')), (2, h_2(i')), \dots, (r-1, h_{r-1}(i'))$

EASY: $F[i] \leq \tilde{F}[i]$

"garbage" due to collisions
with other items i'

each cell $T[j][h_j(i)] = F[i] + X_{ji}$

Since we only increment counters $\Rightarrow X_{ji} \geq 0$

$\Rightarrow \tilde{F}[i] = \min_{j \in [r]} T[j][h_j(i)] \geq F[i]$

$\tilde{F}[i] \leq F[i] + \epsilon \|F\|$ with probability $> 1 - \delta$

Good news is impressively low space: $O(\epsilon^{-1} \log^{-1})$ words

$$\blacktriangleright \boxed{E[I_{j \neq i}]} = \underbrace{\Pr[I_{j \neq i} = 1]}_{\text{UNION BOUND}} = \boxed{\frac{\varepsilon}{c}}$$

$$\hookrightarrow \Pr[\exists v \in [c] : k \neq i \wedge h_j(k) = v \wedge h_j(i) = v] \stackrel{\text{UNION BOUND}}{=} \sum_{v \in [c]} \Pr[k \neq i \wedge h_j(k) = v \wedge h_j(i) = v] =$$

h_j is 2-wise independent

$$\sum_{v \in [c]} \underbrace{\Pr[h_j(i) = v]}_{\frac{1}{c}} \cdot \underbrace{\Pr[h_j(k) = v \wedge k \neq i]}_{\sim \frac{1}{c}} \approx$$

$h_j: [n] \rightarrow [c]$

$$\sum_{v \in [c]} \frac{1}{c^2} = \frac{1}{c} = \frac{\varepsilon}{c}$$

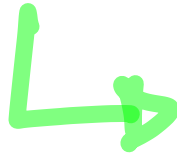
by the choice of $c = \frac{e}{\varepsilon}$

□

$$\blacktriangleright E[X_{ji}] = \sum_{k \in [n]} \underbrace{(E[I_{jik}])}_{\frac{\varepsilon}{e}} \cdot F[k] = \frac{\varepsilon}{e} \underbrace{\sum_{k \in [n]} F[k]}_{\|F\|} = \frac{\varepsilon}{e} \|F\| \quad \square$$

Hence

$$\boxed{\varepsilon \|F\| = e E[X_{ji}]} \quad (*)$$



Recall Markov's inequality

$$\Pr[X > a] \leq \frac{E[X]}{a}$$

Bound error probability by δ :

$$\Pr[\tilde{F}[i] > F[i] + \epsilon \|F\|] =$$

$\tilde{F}[i]$ is a min operation $\Rightarrow$ this holds for any j

$$\Pr[\forall \text{row } j \in [r]: \cancel{F[i]} + X_{ji} > \cancel{F[i]} + \underbrace{\epsilon \|F\|}_{= e E[X_{ji}]}] =$$

$$\Pr[\forall \text{row } j \in [r]: X_{ji} > e E[X_{ji}]] =$$

$h_0, h_1, \dots, h_{r-1}$ are independently chosen

$$\prod_{j=0}^{r-1} \Pr[X_{ji} > \underbrace{e E[X_{ji}]}] < \text{MI} \prod_{j=0}^{r-1} \frac{\cancel{E[X_{ji}]} }{\underbrace{e E[X_{ji}]} } = \left(\frac{1}{e}\right)^r = \delta$$

as $r = \ln \delta^{-1}$

